



UNIVERSITY OF MINES AND TECHNOLOGY, TARKWA

SECOND SEMESTER EXAMINATION, APRIL 2026

COURSE NO : 361

COURSE NAME: PROBABILITY AND STATISTICS

CLASS: CE/EL/MC III

TIME: 3 HOURS

Name: _____

Index Number: _____

SECTION A

Answer all objective questions using the provided MCQ answer sheet. Clearly and neatly shade the correct option.

1. The resistance values (in ohms) of 8 resistors are measured as:

10, 12, 11, 13, 10, 14, 10, 80

A technician reports the mean resistance as the best summary. Why is this misleading?

- | | |
|--|--|
| a. The mean cannot be computed for resistance values | c. The outlier value of $80\ \Omega$ inflates the mean, making it unrepresentative |
| b. The mean is always smaller than the median | d. Resistance values must follow a normal distribution before using the mean |
2. A mechanical engineer measures the diameter of 100 pistons produced daily over 20 days. She wants to compare variability between two production lines that produce pistons of different mean diameters. Which measure should she use?
- | | |
|---|--|
| a. Standard deviation, because it measures spread | c. Coefficient of variation, because it measures relative spread |
| b. Range, because it captures extreme values | d. Variance, because it squares the deviations |
3. An electrical engineer records the voltage output of a power supply every 5 minutes over 8 hours. The first reading was taken at minute 3. This is an example of:
- | | |
|---------------------------|------------------------|
| a. Stratified sampling | c. Systematic sampling |
| b. Simple random sampling | d. Cluster sampling |

4. A dataset of engine temperatures is described by $\bar{x} = 200^{\circ}\text{C}$ and $s = 40^{\circ}\text{C}$. A second engine dataset has $\bar{x} = 400^{\circ}\text{C}$ and $s = 40^{\circ}\text{C}$. Which engine shows greater relative variability?
- Both engines, since $s = 40$ for both
 - The first engine, since its CV = 20% compared to CV = 10%
 - The second engine, since its mean is higher
 - Neither, since standard deviation is the only measure of variability
5. A factory has 4 production shifts. A quality engineer selects 30 components from each shift to test for defects. Which sampling method is this, and what is its key advantage?
- Systematic sampling; it selects items at fixed intervals
 - Stratified sampling; it ensures each shift is represented
 - Cluster sampling; it groups components by shift
 - Convenience sampling; it is the easiest method
6. An electrical circuit has two independent components A and B with $P(\text{A fails}) = 0.05$ and $P(\text{B fails}) = 0.08$. If the circuit fails only when **both** components fail, what is the probability of circuit failure?
- 0.130
 - 0.004
 - 0.076
 - 0.126
7. A machine has two independent warning lights, R (red) and Y (yellow), with $P(R) = 0.10$ and $P(Y) = 0.15$. What is the probability that **at least one** light activates?
- 0.25
 - 0.015
 - 0.235
 - 0.765
8. A batch of 500 capacitors contains 50 defective ones. Using Bayes' theorem: a supplier delivers 60% of capacitors from Factory X (defect rate 5%) and 40% from Factory Y (defect rate 15%). Given a randomly selected capacitor is defective, what is the probability it came from Factory Y?
- 0.400
 - 0.667
 - 0.333
 - 0.500
9. A Bernoulli trial models a **single** test of a transistor where it either passes ($p = 0.92$) or fails ($1 - p = 0.08$). What is the variance of this Bernoulli random variable?
- 0.92
 - 0.08
 - 0.0736
 - 0.8464

10. A quality inspector checks mechanical valves where each valve independently has a 3% chance of being faulty. In a batch of 25 valves, what is the expected number of faulty valves and its variance?
- a. $E = 0.75$, $\text{Var} = 0.7275$ c. $E = 0.75$, $\text{Var} = 0.03$
- b. $E = 3$, $\text{Var} = 0.75$ d. $E = 25$, $\text{Var} = 0.7275$
11. A software team is testing modules independently, each with a 20% chance of containing a critical bug. What is the probability that the first critical bug is found on the 4th module tested?
- a. $(0.20)^4$ c. $(0.80)^3 \times (0.20)$
- b. $\binom{4}{1}(0.20)^1(0.80)^3$ d. $(0.20)^3 \times (0.80)$
12. An electrical engineer tests circuit boards one at a time until finding the second defective board. Each board independently has a 10% defect rate. Which distribution models the number of boards tested?
- a. Binomial distribution c. Negative binomial distribution
- b. Geometric distribution d. Poisson distribution
13. A mechanical pump breaks down on average 2 times per month. Using the Poisson distribution, what is the probability it breaks down **exactly 0 times** in a given month?
- a. $2e^{-2}$ c. e^{-2}
- b. $1 - e^{-2}$ d. e^2
14. Server request arrivals follow a Poisson distribution with $\lambda = 4$ per minute. What is the probability of receiving more than 1 request in a given minute?
- a. $1 - e^{-4}$ c. $1 - e^{-4}(1 + 4)$
- b. $e^{-4}(1 + 4)$ d. $4e^{-4}$
15. The load (in kN) on a structural beam is uniformly distributed between 10 and 30 kN, i.e. $X \sim U(10, 30)$. What is $P(15 \leq X \leq 25)$?
- a. 0.25 b. 0.75 c. 0.50 d. 1.00

16. The time between successive failures of a cooling fan is uniformly distributed between 200 and 600 hours, i.e. $X \sim U(200, 600)$. What is the standard deviation of X ?
- a. 200 b. 400 c. 115.47 d. 133.33
17. The tensile strength of steel rods follows $X \sim N(500, 20^2)$ MPa. A rod is **rejected** if its strength is below 460 MPa. What is the standardised Z -value for the rejection threshold?
- a. $Z = 2$ b. $Z = -2$ c. $Z = -0.2$ d. $Z = 0.2$
18. The output voltage of a power supply follows $X \sim N(12, 0.25)$ V. What is the probability the voltage exceeds 12.5 V? (*Use: $P(Z < 1) = 0.8413$*)
- a. 0.8413 b. 0.3413 c. 0.1587 d. 0.6587
19. CPU processing times follow $X \sim N(300, 25^2)$ ms. What is the probability a task takes between 275 ms and 350 ms? (*Use: $P(Z < -1) = 0.1587$; $P(Z < 2) = 0.9772$*)
- a. 0.8185 b. 0.7745 c. 0.9772 d. 0.1587
20. The diameter of machined shafts follows $N(50, 0.04)$ mm. Shafts are accepted if the diameter lies between 49.6 mm and 50.4 mm. What proportion of shafts are accepted? (*Use: $P(Z < 2) = 0.9772$; $P(Z < -2) = 0.0228$*)
- a. 0.9772 b. 0.0456 c. 0.9544 d. 0.8413
21. The number of power outages X per week at a data centre has the following distribution:

X	0	1	2	3
$P(X)$	0.50	0.30	0.15	0.05

What is the expected number of outages per week?

- a. 1.00 b. 0.75 c. 0.50 d. 1.50
22. A continuous random variable X has pdf $f(x) = \frac{1}{8}x$, $0 \leq x \leq 4$. Compute $E(X)$.
- a. 1.0 b. 2.0 c. $\frac{8}{3}$ d. 4.0

23. Using $\text{Var}(X) = E(X^2) - [E(X)]^2$: a current signal X has $E(X) = 3$ A and $E(X^2) = 13$ A². What is $\text{Var}(X)$?
- a. 10 b. 4 c. 9 d. 16
24. A maintenance cost function is $C = 500X + 200$, where X is the number of machine breakdowns with $E(X) = 3$ and $\text{Var}(X) = 4$. Find $E(C)$ and $\text{Var}(C)$.
- a. $E(C) = 1700$, $\text{Var}(C) = 2200$ c. $E(C) = 1700$, $\text{Var}(C) = 1,000,000$
- b. $E(C) = 1500$, $\text{Var}(C) = 1,000,000$ d. $E(C) = 1700$, $\text{Var}(C) = 4$
25. Two independent machines produce components. Machine A has $E(X_A) = 50$ units and $\text{Var}(X_A) = 9$. Machine B has $E(X_B) = 40$ units and $\text{Var}(X_B) = 16$. What is $\text{Var}(X_A + X_B)$?
- a. 7 b. 25 c. 144 d. 625
26. The pdf of a random variable X is $f(x) = 3x^2$, $0 \leq x \leq 1$. Verify which value is $E(X^2)$:
- a. $\frac{3}{4}$ b. $\frac{3}{5}$ c. $\frac{1}{2}$ d. $\frac{9}{16}$
27. A sample of 50 light bulbs from a production line gives a mean lifetime of $\bar{x} = 1200$ hours with $\sigma = 100$ hours. Construct a **95%** confidence interval for the true mean lifetime. (*Use $z_{0.025} = 1.96$*)
- a. 1200 ± 27.72 hours c. 1200 ± 196 hours
- b. 1200 ± 1.96 hours d. 1200 ± 14.14 hours
28. An engineer constructs a 90% CI as $[48.5, 51.5]$ volts and a 99% CI from the **same data** as $[47.1, 52.9]$ volts. A colleague says: “*The 90% CI is wrong because it is narrower.*” Is this correct?
- a. Yes; a wider interval is always more accurate c. No; higher confidence requires a wider interval, so both are valid
- b. Yes; the 90% CI must be wider than the 99% CI d. No; both intervals must always have the same width

29. A manufacturer wants to estimate the mean breaking strength of cables to within ± 10 N with 95% confidence. Given $\sigma = 40$ N and $z_{0.025} = 1.96$, what is the minimum sample size required?
- a. 8 b. 62 c. 61 d. 16
30. A 95% confidence interval for the mean current draw of a motor is computed as $[4.8, 5.6]$ A. Which interpretation is correct?
- a. There is a 95% probability the true mean is between 4.8 A and 5.6 A c. If many samples were taken, 95% of such intervals would contain the true mean
- b. 95% of all motors draw between 4.8 A and 5.6 A d. The sample mean equals the true mean with 95% accuracy
31. A quality engineer wants to reduce the width of a confidence interval by half while keeping the confidence level fixed. What must happen to the sample size?
- a. It must double c. It must quadruple
- b. It must be halved d. It must increase by a factor of $\sqrt{2}$
32. A motor manufacturer claims the mean operating temperature is 75°C . A test engineer samples 40 motors and finds $\bar{x} = 78^\circ\text{C}$ with $\sigma = 6^\circ\text{C}$. The correct test statistic is:
- a. $z = \frac{78 - 75}{6}$ c. $z = \frac{78 - 75}{6/\sqrt{40}}$
- b. $z = \frac{75 - 78}{6/\sqrt{40}}$ d. $z = \frac{78 - 75}{6 \times \sqrt{40}}$
33. A software company claims its application uses on average 512 MB of RAM. A developer tests 36 runs and finds $\bar{x} = 530$ MB with $\sigma = 60$ MB. At $\alpha = 0.05$ (two-tailed, $z_{0.025} = 1.96$), what is the conclusion?
- a. Fail to reject H_0 ; the test statistic is $z = 1.80 < 1.96$ c. Reject H_0 ; the test statistic is $z = 1.80 > 1.645$
- b. Reject H_0 ; the test statistic is $z = 2.00 > 1.96$ d. Accept H_0 ; 512 MB is the correct mean

34. An electrical engineer tests whether the mean voltage of a power supply differs from 220 V. A sample of $n = 64$ gives $\bar{x} = 217$ V with $\sigma = 8$ V. Which of the following is the correct hypothesis setup and decision at $\alpha = 0.05$? (*Use* $z_{0.025} = 1.96$)
- a. $H_0 : \mu = 220$ vs $H_a : \mu \neq 220$; $z = -3.0$; reject H_0
- b. $H_0 : \mu = 220$ vs $H_a : \mu \neq 220$; $z = -3.0$; fail to reject H_0
- c. $H_0 : \mu \neq 220$ vs $H_a : \mu = 220$; $z = -3.0$; reject H_0
- d. $H_0 : \mu = 220$ vs $H_a : \mu < 220$; $z = 3.0$; reject H_0
35. A mechanical engineer claims a new lubricant increases mean engine life above 5000 hours. A sample of 49 engines gives $\bar{x} = 5150$ hours with $\sigma = 350$ hours. At $\alpha = 0.05$ (one-tailed, $z_{0.05} = 1.645$), what is the correct conclusion?
- a. Fail to reject H_0 ; $z = 3.0 > 1.645$
- b. Reject H_0 ; $z = 1.50 < 1.645$
- c. Reject H_0 ; $z = 3.0 > 1.645$; evidence supports the claim
- d. Fail to reject H_0 ; $z = 1.50 < 1.645$
36. In hypothesis testing, a Type I error in the context of a circuit quality test means:
- a. Accepting a defective batch as good
- b. Accepting a good batch as defective
- c. Rejecting a good batch as defective
- d. Failing to test the batch at all
37. A researcher obtains $p\text{-value} = 0.03$ in a two-tailed test at $\alpha = 0.05$. A colleague argues: “*This proves H_0 is false.*” Evaluate this statement.
- a. Correct; $p < \alpha$ mathematically proves H_0 false
- b. Correct; the p -value is below 0.05 confirming H_a
- c. Incorrect; we *reject* H_0 but retain a probability α of a Type I error
- d. Incorrect; we can only reject H_0 when $p\text{-value} = 0$
38. From 10 data points on engine speed x (rpm) and fuel consumption y (L/hr), a regression yields: $\sum(x_i - \bar{x})(y_i - \bar{y}) = 360$ and $\sum(x_i - \bar{x})^2 = 120$. What is the estimated slope b_1 ?
- a. 0.33
- b. 3.0
- c. 43200
- d. 240
39. A regression model for predicting CPU temperature (y , °C) from processor load (x , %) gives $\hat{y} = 40 + 0.5x$. A student interprets the slope as: “*When load increases by 1%, temperature increases by 0.5° C on average.*” Is this interpretation correct?

- a. No; the slope represents the value of y when $x = 0$
- b. No; the slope must be multiplied by $\bar{x}$ first
- c. Yes; the slope gives the average change in y per unit increase in x
- d. No; the slope represents total variability in y

40. A regression analysis on voltage x and power output y of a generator gives $SST = 500$ and $SSE = 125$. A technician claims: “The predictor variable x explains 75% of the variability in power output.” Verify this using R^2 .

- a. Incorrect; $R^2 = SSE/SST = 0.25$
- b. Incorrect; $R^2 = SST - SSE = 375$
- c. Correct; $R^2 = 1 - SSE/SST = 0.75$
- d. Correct; $R^2 = SSE/SST \times 100 = 75\%$

SECTION B

Carefully read each question and attempt only ONE (1) question

Question 1

- (a) A hospital administrator has monitored daily emergency admissions over several years and proposed that the data follow a Poisson distribution with a historical mean of $\lambda = 3$ admissions per day. Before acting on this assumption, the administrator must critically evaluate the model and use it to inform hospital resource planning.
- i. State two conditions that must be satisfied for the Poisson distribution to be a valid model for daily emergency admissions. Assess whether these conditions are likely to hold in a real hospital setting.
 - ii. Using the Poisson model with $\lambda = 3$, compute the probability that exactly 2 emergency admissions occur on a given day. Interpret this result in the context of hospital staffing decisions.
 - iii. Find the probability that no emergency admissions occur on a particular day. The hospital administrator uses this result to justify reducing the overnight staff on low-risk days. Critically evaluate whether this is a statistically sound decision.
 - iv. Compute the probability that either 3 or 4 emergency admissions occur on a given day. Using your answer, determine the probability that the number of admissions falls outside this range, and comment on what this means for resource allocation.
 - v. The administrator notices that emergency admissions tend to spike significantly on weekends compared to weekdays. What does this observation suggest about the suitability of using a single Poisson model for the entire week? Propose a more appropriate modelling approach.

Question 2

- (a) A solid propellant powers the aircrew escape system. The burning rate of this propellant is a crucial characteristic of the product. The specifications require that the mean burning rate must be 50 centimeters per second. We know that the standard deviation of the burning rate is 2 centimeters per second. The experimenter specifies a significance level of $\alpha = 0.05$, selects a random sample of 25 and obtains a sample average burning rate of 51.3 centimeters per second.
- Clearly state the null and alternative hypotheses for this test, identifying whether it is a one-tailed or two-tailed test.
 - Compute the test statistic and identify the critical region at $\alpha = 0.05$
 - What conclusions should be drawn? Justify your answer statistically.
 - Suppose the true mean burning rate is actually 50.5 cm/s. What type of error, if any, might the original test have committed? Explain.
- (b) An engineer claims: "If two events are independent, then they cannot occur together."
- State whether this claim is true or false.
 - Justify your answer with a counterexample.

Question 3

- (a) The tensile strength (MPa) of steel rods produced by an automated manufacturing process is modelled as a Normal random variable with mean $\mu = 520$ MPa and standard deviation $\sigma = 18$ MPa. A rod is *rejected* if its tensile strength falls below 490 MPa or exceeds 560 MPa. A quality-control engineer randomly samples $n = 36$ rods from a production batch.
- Show that the proportion of rods rejected due to *low* tensile strength is approximately 4.78%. State clearly the z -value used.
 - Find the probability that a randomly chosen rod has tensile strength between 490 MPa and 560 MPa, i.e. find $P(490 < X < 560)$.
 - The manufacturer wishes to reset the upper rejection limit c such that only 2% of rods are rejected for exceeding it, that is $P(X > c) = 0.02$. Find the value of c to one decimal place.
 - For the random sample of $n = 36$ rods, let $\bar{X}$ denote the sample mean tensile strength.
- (b) A fair coin is tossed three times. Let represent the number of heads which come up. Find the cumulative distribution function and probability mass function.

— END OF EXAMINATION —

Descriptive Statistics	
Mean (Average)	$\mu = \frac{\sum X_i}{n}$
Variance	$\sigma^2 = \frac{\sum (X_i - \mu)^2}{n}$
Standard Deviation	$\sigma = \sqrt{\frac{\sum (X_i - \mu)^2}{n}}$
Probability Rules	
Addition Rule	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Multiplication Rule	$P(A \cap B) = P(A) P(B A) = P(B) \cdot P(A B)$
Bayes' Theorem	$P(A B) = \frac{P(B A) P(A)}{P(B)}$
Discrete Distributions	
Binomial	$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$
Poisson	$P(X = k) = \frac{e^{-\lambda} \cdot \lambda^k}{k!}$
Negative Binomial	$P(X = k) = \binom{k-1}{n-1} p^n (1 - p)^{k-n}$
Bernoulli	$P(X = k) = p^k (1 - p)^{1-k}$
Geometric	$P(X = k) = (1 - p)^{k-1} \cdot p \quad P(X \leq k) = 1 - (1 - p)^k$
Continuous Distributions	
Normal	$f(x) = \frac{1}{\sigma \sqrt{2\pi}} \cdot e^{-\frac{1}{2} \left(\frac{x - \mu}{\sigma} \right)^2}$
Uniform	$f(x) = \frac{1}{b - a} \text{ for } a \leq x \leq b$
Standard Normal (Z-score)	$Z = \frac{X - \mu}{\sigma}$
Confidence Intervals & Hypothesis Testing	
CI for μ (σ known)	$CI = \bar{x} \pm z \cdot \frac{\sigma}{\sqrt{n}}$
CI for μ (σ unknown)	$CI = \bar{x} \pm t_{\alpha/2, n-1} \cdot \frac{s}{\sqrt{n}}$
Test Statistic for Mean	$Z = \frac{X - \mu_0}{\sigma / \sqrt{n}}$